

Harshit Singh

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EDUCATION

Vellore Institute of Technology, Chennai

B.Tech, Electronics and Computer Engineering — CGPA: 8.07

Chennai, India

2023 – 2027

New Era School (CBSE)

Class XII: 87.2% — Class X: 89.9%

Ghaziabad, Uttar Pradesh

2020 – 2023

EXPERIENCE

AI & Robotics Intern — Autonomous Mobile Robotics [Certificate]

Jul 2025 – Aug 2025

STEMTEC AI & Robotics Technology Pvt. Ltd.

Chennai, India

- Developed a **ROS 2 Jazzy reactive navigation node** processing /scan LiDAR data at **10 Hz**, enabling real-time obstacle avoidance within a **0.75 m safety radius** and reducing collision events during indoor navigation tests by **~80%**.
- Implemented timer-driven velocity control nodes publishing to /cmd_vel with **sub-50 ms command latency**, achieving stable motion at **0.4 m/s** and consistent bidirectional odometry validation across simulation and hardware deployments.

PROJECTS

SPARC — Sign Language Processing & Recognition System (OpenCV, Deep Learning) [Demo] [Web-App]

- Developed a real-time ISL/ASL recognition pipeline (A–Z, 0–9, words) trained on **12k+ samples**, achieving **92.4% accuracy** at **15–20 FPS** with end-to-end latency under **180 ms** including TTS.
- Engineered a modular inference architecture using **MediaPipe, TensorFlow/Keras, and OpenCV**, enabling deployment across **3+ configurations** with automatic fallback and **sub-250 ms** recovery from sensor failures.

TRIDENT — Tactical Disaster Response & Rescue System (IoT, Robotics)

[Demo] [GitHub]

- Built an edge-based rescue platform integrating a smart wearable, dashboard, and ROV, streaming **real-time vitals, motion, stress, and GPS data** with alert latency under **1.8 s** and telemetry uptime of **>95%**.
- Designed rule-based and probabilistic scoring to classify victims into **Priority 1–5**, reducing manual triage time by **40%** and improving decision accuracy by **30%**.

DrishtiGuide — Assistive Navigation System (YOLOv8, IoT)

[GitHub]

- Built a real-time multimodal navigation pipeline (audio, haptic, ultrasonic), reducing missed-obstacle events by **60%** versus cane-only navigation in field trials.
- Integrated **YOLOv8, OpenCV, and PyTorch** with ESP8266 telemetry, achieving **sub-120 ms inference latency** and **>99%** reliable GPS alert delivery to caregivers.

SKILLS

Programming Languages: C, Python, Java, HTML, CSS

Frameworks & Libraries: OpenCV, PyTorch, TensorFlow, YOLOv8, ROS 2, NumPy, Pandas, MediaPipe, CVZone

Systems & Platforms: Linux, ROS 2 Middleware, Real-time Data Pipelines

Development Tools: Visual Studio Code, Git, Github

Professional Skills: Software Design, Problem-solving, Debugging, Collaboration, Time Management

ACHIEVEMENTS & LEADERSHIP

- SIH'25 Finalist** — advanced to **Top 50 out of 864 teams** internally and **Top 5 out of 500 teams** nationally for the Grand Finale. [Certificate]
- 3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; ₹ 15,000 award. [Certificate]
- Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35 teams**; ₹ 5,000 award.
- Deputy Operations Lead, Team Namu Nirvana (VIT Chennai)** — led **100+ member teams** across **5–6 competitions/semester**, boosting execution efficiency by **35%** and securing **2 sponsors**. [LOA]

LANGUAGES

English (Fluent) — Hindi (Native)